

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250286573

Kind Code

A1

Publication Date

September 11, 2025

Inventor(s)

MERLICS; STEPHAN et al.

COMMUNICATIONS DEVICE WITH BROADBAND TRANSPORT AND RELATED METHODS

Abstract

A communications device includes a portable housing, a cellular transceiver carried by the portable housing and configured to connect with an external cellular network, a WLAN transceiver carried by the portable housing and configured to connect with a local wireless device, and a geolocation receiver carried by the portable housing and configured to generate geolocation data. The communications device also includes RF antennas coupled to the cellular transceiver, the WLAN transceiver, and the geolocation receiver, a wired port carried by the portable housing, and a processor carried by the portable housing. The processor is configured to route communications from the wireless device to the external cellular network.

Inventors: MERLICS; STEPHAN (NICEVILLE, FL), MITCHELL-HUDDLE; MARK (VIRGINIA BEACH, VA), AXELSON; MATTHEW (VIRGINIA BEACH, VA)

Applicant: Aegis Terra LLC. (Sheridan, WY)

Family ID: 96950142

Appl. No.: 19/072887

Filed: March 06, 2025

Related U.S. Application Data

us-provisional-application US 63561875 20240306

Publication Classification

Int. Cl.: H04B1/3827 (20150101); H04B1/00 (20060101); H04L45/85 (20220101); H04W4/029 (20180101); H04W84/12 (20090101)

U.S. Cl.:

Background/Summary

RELATED APPLICATION [0001] This application is based upon prior filed copending Application No. 63/561,875 filed Mar. 6, 2024, the entire subject matter of which is incorporated herein by reference in its entirety.

TECHNICAL FIELD

[0002] The present disclosure relates to the field of communications, and, more particularly, to a communications device and related methods.

BACKGROUND

[0003] Mobile communications devices have become an integral part of society over the last two decades. Indeed, more than eighty-two percent of Americans own a mobile communications device, for example, a cell phone. The typical mobile communications device includes an antenna, and a transceiver coupled to the antenna. The transceiver and the antenna cooperate to transmit and receive communication signals. In some applications, communications devices with proprietary waveforms are deployed, so called software radios. Due to the nature of these devices, they are limited to geographic areas with compatible infrastructure.

SUMMARY

[0004] Generally, a communications device comprises a portable housing, a cellular transceiver carried by the portable housing and configured to connect with an external cellular network, a wireless local area network (WLAN) transceiver carried by the portable housing and configured to connect with at least one local wireless device, and a geolocation receiver carried by the portable housing and configured to generate geolocation data. The communications device also includes a plurality of radio frequency (RF) antennas coupled to the cellular transceiver, the WLAN transceiver, and the geolocation receiver, at least one wired port carried by the portable housing, and a processor carried by the portable housing and coupled to the cellular transceiver, the WLAN transceiver, the geolocation receiver, and the at least one wired port. The processor is configured to route communications from the at least one local wireless device to the external cellular network.

[0005] More specifically, the communications device may also include an RF diplexer coupled between the plurality of RF antennas and the cellular transceiver, the WLAN transceiver, and the geolocation receiver.

[0006] Another aspect is directed to a method for operating a communications device comprising a portable housing, a cellular transceiver carried by the portable housing and configured to connect with an external cellular network, and a WLAN transceiver carried by the portable housing and configured to connect with at least one local wireless device. The communications device also includes a geolocation receiver carried by the portable housing and configured to generate geolocation data, a plurality of RF antennas coupled to the cellular transceiver, the WLAN transceiver, and the geolocation receiver, and at least one wired port carried by the portable housing. The method comprises operating a processor coupled to the cellular transceiver, the WLAN transceiver, the geolocation receiver, and the at least one wired port. The processor is configured to route communications from the at least one local wireless device to the external cellular network.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0007] FIG. **1** is a schematic diagram of a communication system, according to the present disclosure.

[0008] FIG. **2** is a circuit diagram of an ethernet controller in an example embodiment of the communications device, according to the present disclosure.

[0009] FIG. **3** is a circuit diagram of an oscillator circuit in an example embodiment of the communications device, according to the present disclosure.

[0010] FIG. **4** is a circuit diagram of a power circuit in an example embodiment of the communications device, according to the present disclosure.

[0011] FIG. **5** is a circuit diagram of a flash memory circuit in an example embodiment of the communications device, according to the present disclosure.

[0012] FIG. **6** is a circuit diagram of a reset circuit in an example embodiment of the communications device, according to the present disclosure.

[0013] FIG. **7** is a circuit diagram of an external wake-up circuit in an example embodiment of the communications device, according to the present disclosure.

[0014] FIG. **8** is a circuit diagram of an ethernet circuit in an example embodiment of the communications device, according to the present disclosure.

[0015] FIG. **9** is a circuit diagram of a USB/power circuit in an example embodiment of the communications device, according to the present disclosure.

[0016] FIGS. **10-11** are circuit diagrams of an RF diplexer in an example embodiment of the communications device, according to the present disclosure.

DETAILED DESCRIPTION

[0017] The present disclosure will now be described more fully hereinafter with reference to the accompanying drawings, in which several embodiments of the invention are shown. This present disclosure may, however, be embodied in many different forms and should not be construed as limited to the embodiments set forth herein. Rather, these embodiments are provided so that this disclosure will be thorough and complete, and will fully convey the scope of the present disclosure to those skilled in the art. Like numbers refer to like elements throughout, and base 100 reference numerals are used to indicate similar elements in alternative embodiments.

[0018] With software defined radio, there can be a problem due to the lack of interoperability with existing widely deployed networks. In remote areas, this issue can be exacerbated. The present disclosure provides an approach to this issue with a Field Broadband Transport (FBT) device. In short, the FBT device provides a conduit between software defined radios and typical wide spread networks.

[0019] In particular, the FBT device may provide a small form factor, ruggedized, body-worn device that provides long-range WiFi, LTE and WLAN/wired connectivity to any edge computing device, Field Compute Device-Wearable (FCD-W) configuration or Manet radio network. Further, single or multiuser connectivity for Line-of-Sight (LoS) local networks may provide users with the ability to transmit and receive vital data from the far-flung deployment areas.

[0020] Referring now to FIG. **1**, a communication system **100** according to the present disclosure is now described. The communication system **100** includes a communications device **101** (i.e., the FBT device), an external cellular network **102** (e.g., 3G-5G, LTE network), a plurality of local wireless devices (LWDs) **103a-103n** (e.g., software defined radios, typical consumer mobile devices, tablet computing devices, etc.), and a plurality of wired devices **104a-104n**.

[0021] The communications device **101** illustratively includes a portable housing **105**. The portable housing **105** may comprise a an IP67 specification housing, which provides for a rugged and durable housing. Further, the portable housing **105** may provide an integrated scratch resistant, high heat tolerant, and salt-water resistant sealed enclosure.

[0022] The communications device **101** illustratively includes a cellular transceiver **106** (e.g., 3G-

5G, LTE network) carried by the portable housing **105** and configured to connect with the external cellular network **102**. In some embodiments, the cellular transceiver **106** may comprise an LTE transceiver, which supports global LTE Bands with 3G/HSPA Fallback, and Verizon Airborne LTE Operations (ALO) Bands for unmanned operations. The cellular transceiver **106** may be agnostic to Global Cellular Carriers and supports private LTE networks.

[0023] The communications device **101** illustratively includes a WLAN transceiver **107** carried by the portable housing **105** and configured to connect with the plurality of LWDs **103a-103n**. In some embodiments, the WLAN transceiver **107** comprises one or more of a WiFi transceiver (i.e. 802.11x transceiver), a Bluetooth transceiver, or a WiMAX transceiver. For example, the WLAN transceiver **107** may comprise a WiFi transceiver having 2.4 GHZ 2×2 MIMO, Maximal Ratio Combining (MRC), Maximal Likelihood (ML) decoding and Low-Density Parity Check (LDPC), which may provide robust RF performance in the most constrained environments.

[0024] The communications device **101** illustratively includes a geolocation receiver **110** carried by the portable housing and configured to generate geolocation data. In some embodiments, the geolocation receiver **110** may comprise a global positioning system (GPS) receiver, for example. The geolocation receiver **110** may be a standalone GPS receiver for asset tracking report formats and timing intervals.

[0025] The communications device **101** illustratively includes a plurality of RF antennas **111a-111n** coupled to the cellular transceiver **106**, the WLAN transceiver **107**, and the geolocation receiver **110**. The communications device **101** comprises one or more wired ports **112** (illustrated as one for drawing clarity) carried by the portable housing **105**, and a processor **113** carried by the portable housing and coupled to the cellular transceiver **106**, the WLAN transceiver **107**, the geolocation receiver **110**, and the wired port **112**.

[0026] The processor **113** is configured to route communications from the plurality of LWDs **103a-103n** to the external cellular network **102**. As will be appreciated, the processor **113** servers as a local router for the plurality of LWDs **103a-103n** and the plurality of wired devices **104a-104n**, exchanging local traffic, and accepting traffic for external transmission via the cellular transceiver **106**.

[0027] The wired port **112** may comprise one or more ports in some embodiments for connection to one or more of the wired devices **104**. For example, the wired port **112** may comprise one or more of an ODU multi-pin port, a USB port, an ethernet communications port, a serial port, or an external power port for operation or charging on-board battery (not shown). Further, the communications device **101** may have fully user configurable networking options, and the communications device may provide secure VPN servers and client capability.

[0028] The communications device **101** illustratively includes an RF diplexer **114** coupled between the plurality of RF antennas **111a-111n** and the cellular transceiver **106**, the WLAN transceiver **107**, and the geolocation receiver **110**. As will be appreciated, the RF diplexer **114** routes received and transmitted signals to the appropriate RF antenna. In some embodiments, the processor **113**, the cellular transceiver **106**, the WLAN transceiver **107**, the geolocation receiver **110**, and the wired port **112** may be provided on an application specific integrated circuit (ASIC). For example, some embodiments may use the pX2LTE-4GL, as available from MicroHard Systems Inc. of Calgary, Canada.

[0029] Another aspect is directed to a method for operating a communications device **101** comprising a portable housing **105**, a cellular transceiver **106** carried by the portable housing and configured to connect with an external cellular network **102**, and a WLAN transceiver **107** carried by the portable housing and configured to connect with a plurality of LWDs **103a-103n**. The communications device **101** also includes a geolocation receiver **110** carried by the portable housing **105** and configured to generate geolocation data, a plurality of RF antennas **111a-111n** coupled to the cellular transceiver **106**, the WLAN transceiver **107**, and the geolocation receiver **110**, and a wired port **112** carried by the portable housing. The method comprises operating a

processor **113** coupled to the cellular transceiver **106**, the WLAN transceiver **107**, the geolocation receiver **110**, and the wired port **112**. The processor **113** is configured to route communications from the plurality of LWDs **103a-103n** to the external cellular network **102**.

[0030] Referring now additionally to FIG. 2, an ethernet controller **1000** to be coupled to the wired port **112** is shown. For example, the ethernet controller **1000** may comprise an AX88179A controller, as available from the ASIX Electronics Corporation of Taiwan.

[0031] Referring now additionally to FIGS. 3-5, an oscillator circuit **1100**, a power circuit **1200**, and an SPI flash circuit **1300** (i.e., a memory circuit) are shown. Each of these circuits would be coupled to the processor **113**. Referring now additionally to FIGS. 6-7, a reset circuit **1400** and an external wakeup circuit **1500** are shown. Each of these circuits would be coupled to the processor **113**. Referring now additionally to FIGS. 8-9, an ethernet circuit **1600** and a USB/power circuit **1700** are shown. Each of these circuits would be coupled to the processor **113**. Referring now additionally to FIGS. 10-11, an RF diplexer **114** is shown.

[0032] Helpfully, the communications device **101** may support multiple connection and configuration options packaged in an IP67, rugged, low power draw solution. This may provide the user with enhanced capabilities, previously requiring multiple devices, now all-in-one scratch resistant, high heat tolerant, salt-water resistant sealed enclosure.

[0033] Many modifications and other embodiments of the present disclosure will come to the mind of one skilled in the art having the benefit of the teachings presented in the foregoing descriptions and the associated drawings. Therefore, it is understood that the present disclosure is not to be limited to the specific embodiments disclosed, and that modifications and embodiments are intended to be included within the scope of the appended claims.

Claims

1. A communications device comprising: a portable housing; a cellular transceiver carried by the portable housing and configured to connect with an external cellular network; a wireless local area network (WLAN) transceiver carried by the portable housing and configured to connect with at least one local wireless device; a geolocation receiver carried by the portable housing and configured to generate geolocation data; a plurality of radio frequency (RF) antennas coupled to the cellular transceiver, the WLAN transceiver, and the geolocation receiver; at least one wired port carried by the portable housing; and a processor carried by the portable housing and coupled to the cellular transceiver, the WLAN transceiver, the geolocation receiver, and the at least one wired port, the processor configured to route communications from the at least one local wireless device to the external cellular network.
 2. The communications device of claim 1 further comprising an RF diplexer coupled between the plurality of RF antennas and the cellular transceiver, the WLAN transceiver, and the geolocation receiver.
 3. A method for operating a communications device comprising a portable housing, a cellular transceiver carried by the portable housing and configured to connect with an external cellular network, a wireless local area network (WLAN) transceiver carried by the portable housing and configured to connect with at least one local wireless device, a geolocation receiver carried by the portable housing and configured to generate geolocation data, a plurality of radio frequency (RF) antennas coupled to the cellular transceiver, the WLAN transceiver, and the geolocation receiver, and at least one wired port carried by the portable housing, the method comprising: operating a processor coupled to the cellular transceiver, the WLAN transceiver, the geolocation receiver, and the at least one wired port, the processor configured to route communications from the at least one local wireless device to the external cellular network.
-